The following pain management protocol is tiered to ensure a global relevance, recognizing that not all analgesic modalities are available to veterinary practitioners and vary from region to region around the world. Its implementation will be guided by the various analgesic modalities available along with the needs of the individual patient requiring treatment. This protocol is reproduced from the WSAVA Pain Committee guidelines, a succinct yet comprehensive review of pain assessment, various pain modalities, and the treatment of various clinically painful scenarios in both dogs and cats. The WSAVA Pain Committee Guidelines are published in the Journal of Small Animal Practice and is available for open access at the Pain Committee pages of **www.wsava.org.**

CANCER PAIN

Pain in cancer patients can be associated with the cancer itself, diagnostic procedures, or treatments, or it can be unrelated to cancer. Pain from cancer itself has varying degrees of severity that is dependent on duration, location, and type of cancer. It may be associated with inflammation, tissue infiltration, mechanical factors (e.g. organ distension), nerve infiltration or compression and potentially factors released by the tumour. Most patients with cancer suffer pain at some point in the course of the disease. The prevalence and severity of pain associated with various cancer types in animals is not well documented.

One of the best documented types of cancer pain is that associated with primary or metastatic bone tumours. Pain results from direct invasion of the bone, microfractures, increased pressure of endosteum, distortion of the periosteum or perilesional inflammation. Another important mechanism is the release of chemical mediators such as amines, peptides, fatty acids, potassium, and prostaglandins. Cancer pain, and bone pain in particular, are often associated with neuropathic-like clinical signs. Dogs with bone cancer can be affected by widespread somatosensory sensitivity and clinical pain is generally refractory to palliative treatment with orally administered analgesics. Cancer treatments including specific chemotherapy agents (chemotherapy induced peripheral neuropathy [CIPN]) and therapeutic radiation (radiation associated pain [RAP]) can be associated with significant pain at the time of treatment, but also for prolonged periods afterwards. Mechanisms of CIPN and RAP are poorly understood and undergoing investigation.

The presence of pain itself (unrelated to cancer) may promote the progression of cancer, and emerging evidence indicates that pre-treatment cancer pain levels may be negatively related to survival. Additionally, evidence indicates that some cancers may coapt (i.e. to bring together) sensory nerves and pain signalling mechanisms to facilitate its progression.

While a detailed understanding of cancer pain mechanisms will eventually lead to specific recommendations, currently a multimodal drug approach to the control of chronic cancer pain is recommended and appropriate management of any non-cancer pain, such as procedure, perioperative pain or other chronic pain conditions such as osteoarthritis (OA), is recommended. In general, for chronic cancer pain, NSAIDs are recommended with the addition of opioids and adjunctive drugs (such as gabapentin) as needed. Other modalities that can prove beneficial are bisphosphonates, chemotherapy, and radiotherapy. Non-drug therapies should be used concurrently. Other forms of adjunctive therapy tend to improve quality of life in cancer patients, although it is not known if they directly induce analgesia.

For additional pharmaceutical dosing information, see the dosing tables in the WSAVA Pain Committee Treatise at www.wsava.org

Monteiro, B.P., Lascelles, B.D.X., Murrell, J., Robertson, S., Steagall, P.V.M. and Wright, B. (2023), 2022 WSAVA guidelines for the recognition, assessment and treatment of pain. J Small Anim Pract, 64: 177-254. https://doi.org/10.1111/jsap.13566

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